System overview: locations and cable bundles

Where each part lives and which cables share each run. Bundles are worked out from the routed 3D model. Circuits on sheets E2–E7 and E9.

Roof platform
PV1 190 W + PV2 130 W + PV5 visor 250 W
ST1 Starlink, U11 visor drive (E9)

internal: PVV

5 CABLES
DO4, PVR, VIS, VSW, VIG

Tent (detachable)
services box: XI1 inlet, Q4 RCBO, XS3 outlet, X5
PV3 + PV4 2 × 100 W flexible (series)

2 CABLES
T1, T2

Engine bay
BT0 start battery + F8 60 A MIDI
FBX fuse box: F9 2 A add-a-fuse (IGN)

Gap junction box
J1–J4 keyed connectors (tent)
glands for the roof cables

2 CABLES
ALT, IGN

9 CABLES
AC4, AC5, DO4, DO5, PVR, PVT
VIS, VSW, VIG

PASSENGER CHASSIS RAIL

13 CABLES
AC4, AC5, DO3, DO4, DO5, ALT
PVR, PVT, IGN, CHB, VIS, VSW
VIG

2 CABLES
ALT, IGN

3 CABLES
DO3, ALT, IGN

12 CABLES
AC4, AC5, DO3, DO4, DO5, ALT
PVR, PVT, IGN, VIS, VSW, VIG

Factory-outlet gland
passenger rear tub wall

1 CABLE
DO3

13 CABLES
AC4, AC5, DO3, DO4, DO5, ALT
PVR, PVT, IGN, CHB, VIS, VSW
VIG

Cabin, under the rear seat
FB3 local fuses
X4 Anderson, X8 12 V, U10 USB-C

Wall drop + tray
drop beside the gland; tray under the lid

14 CABLES
DCF, AC1, AC5, ALT, PVR, PVT
SW1, IGN, CHB, SW2, D1, D2
D4, VIG

13 CABLES
DCF, AC1, AC4, DO3, DO4, DO5
SW1, SW2, D1, D2, D4, VIS
VSW

TUB

Front bay
BT1/BT2, F1/F2, BB1–BB4, QS1, P1, F3
U1 MultiPlus-II, U2 Orion XS, U3/U4 MPPTs
F4–F7 MIDI/MEGA, M1 fan

internal: B1P, B1N, B2P, B2N, RB1, RB2, SH1, SH2, CT1, INVP, INVN, ORO, MPR, MPT, RBF, SHV, FAN, PE1

Rear control cabinet
FB1 fuse block, RCBO enclosure (Q1–Q3)
U5 Cerbo GX, rear panel: U6, S1, S2,
XS1 GPO, X1/X2, U7/U8, X6

internal: AC2, DO1A, DO1B, DO1C, DO1D, CGX, D3

2 CABLES
AC3, DO2

Power slide pod
FB2, XS2 GPO, X3, X7, U9

ROUTING RULES

- Every cable that leaves the tub goes out through the factory 240 V outlet hole (passenger rear wall) and along the passenger chassis rail.
- Nothing passes through the load lid.
- The tent harness (dashed) unplugs at the gap junction, so the tent lifts off with its services box and solar.
- All switching is at the rear panel: battery isolate (S1 → QS1), inverter on/off (S2), GX Touch 50.
- Line weight ∝ number of cables in the run.

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E1 of E9**
TITLE **System overview: locations and cable bundles**	REV / DATE **v0.11 · 2026-10-20**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.11.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

DC power: battery bank and main distribution

Front bay, between the wheel arches. 12.8 V LiFePO$_4$ bank → isolation → protection → system busbars.

FRONT BAY
tub front wall, between the arches (lid can't open with the tent on; everything is serviced from the rear)

RBF · 1.0 mm² → S1 +12 V feed
always live (bank side) · sheet E6

F10 2 A

QS1 Battery switch
Blue Sea ML-RBS 7713
500 A latching, manual knob
IN · OUT · CTRL

SW1 · 4-core 1.0 mm² → S1 rear panel
sheet E6

BB1 Bank + busbar

B1P · 70 mm² → F1 MRBF 300 A → + M8 → **BT1 Battery 1**
iTechworld iTech400X Pro
12.8 V · 400 Ah LiFePO$_4$
stands on end, terminals rear
internal BMS (Bluetooth only)
− M8 → B1N · 70 mm²

B2P · 70 mm² → F2 MRBF 300 A → + M8 → **BT2 Battery 2**
iTechworld iTech400X Pro
12.8 V · 400 Ah LiFePO$_4$
stands on end, terminals rear
internal BMS (Bluetooth only)
− M8 → B2N · 70 mm²

BB2 Bank − busbar

RB1 · 70 mm² → BATT
SH1 · 70 mm²

RB2 · 70 mm² → OUT
SH2 · 70 mm²

BB3 System + busbar

BB4 System − busbar

F11 1 A

SHV · 0.75 mm² → V+

D2 · VE.Direct → U5 Cerbo
sheet E6
VE.D

P1 Battery monitor
Victron SmartShunt 500 A
all − current passes it
BATT · SYS

CT1 · 2×50 mm² → F3 Class-T 400 A → INVP · 2×50 mm² → DC+

U1 Inverter / charger
Victron MultiPlus-II 12/3000/120-32
230 V · 2400 W cont. · 120 A charge
DC: 2 × 50 mm² per pole (0–5 m)
AC-in / AC-out → sheet E5
VE.Bus, remote on/off → sheet E6
lies flat beside the batteries

DC− → INVN · 2×50 mm²

CASE ⏚ → PE1 · 50 mm²

F12 5 A → **M1** fan 92 mm → FAN · 1.5 mm² twin

F4 MIDI 70 A → **ORO+ ← U2 Orion XS**
16 mm² twin · sheet E4

F5 MIDI 60 A → **MPR+ ← U3 MPPT 100/50**
16 mm² twin · sheet E4

F6 MIDI 20 A → **MPT+ ← U4 MPPT 75/15**
6 mm² twin · sheet E4

F7 MEGA 100 A → **DCF+ → FB1**
25 mm² twin · E3

DCF− → FB1
sheet E3

ORO− ← U2 Orion XS (E4)

MPR− ← U3 MPPT 100/50 (E4)

MPT− ← U4 MPPT 75/15 (E4)

CHB · 25 mm² → chassis rail bond, via the gland

BATTERY BANK

- Cut B1P = B2P and B1N = B2N to identical length with identical lugs, so the batteries share current.
- F1/F2 MRBF 300 A sit directly on each + stud.
- 70 mm² tinned, fine-strand; tinned lugs crimped with a hex crimper, adhesive heat-shrink.
- Torque every stud to the maker's figure; re-check after the first 500 km of corrugations.

ISOLATION

- QS1 opens the main + between bank and system. Only the SmartShunt sense (F11) and QS1's own control feed (F10) stay live when it is open.
- QS1 has a manual knob if the control fails.
- Turn the inverter off (S2) before closing QS1: limits the inrush into the MultiPlus capacitors.

EARTHING

- One bond only: BB4 → chassis rail (CHB, 25 mm²).
- MultiPlus case → BB4 (PE1). Victron: same size as the DC negative cable.
- Orion XS is non-isolated: start-battery − and house − are common through ALT (sheet E4).

VICTRON MULTIPLUS-II 12/3000

- F3: Class-T 400 A, Blue Sea 5502 holder.
- CT1 / INVP / INVN: 2 × 50 mm² per pole.
- Keep INVP/INVN together, ≤ 1 m each; they pass the cable window in the board (v0.9).
- MIDI F4–F6 and MEGA F7 on the BB3 studs.

LEGEND · line weight ∝ conductor size · ID · `size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) · DC − (black) · Protective earth (green/yellow) · Data (VE.Bus / VE.Direct / USB) · Control (12 V switched) · Signal / sense

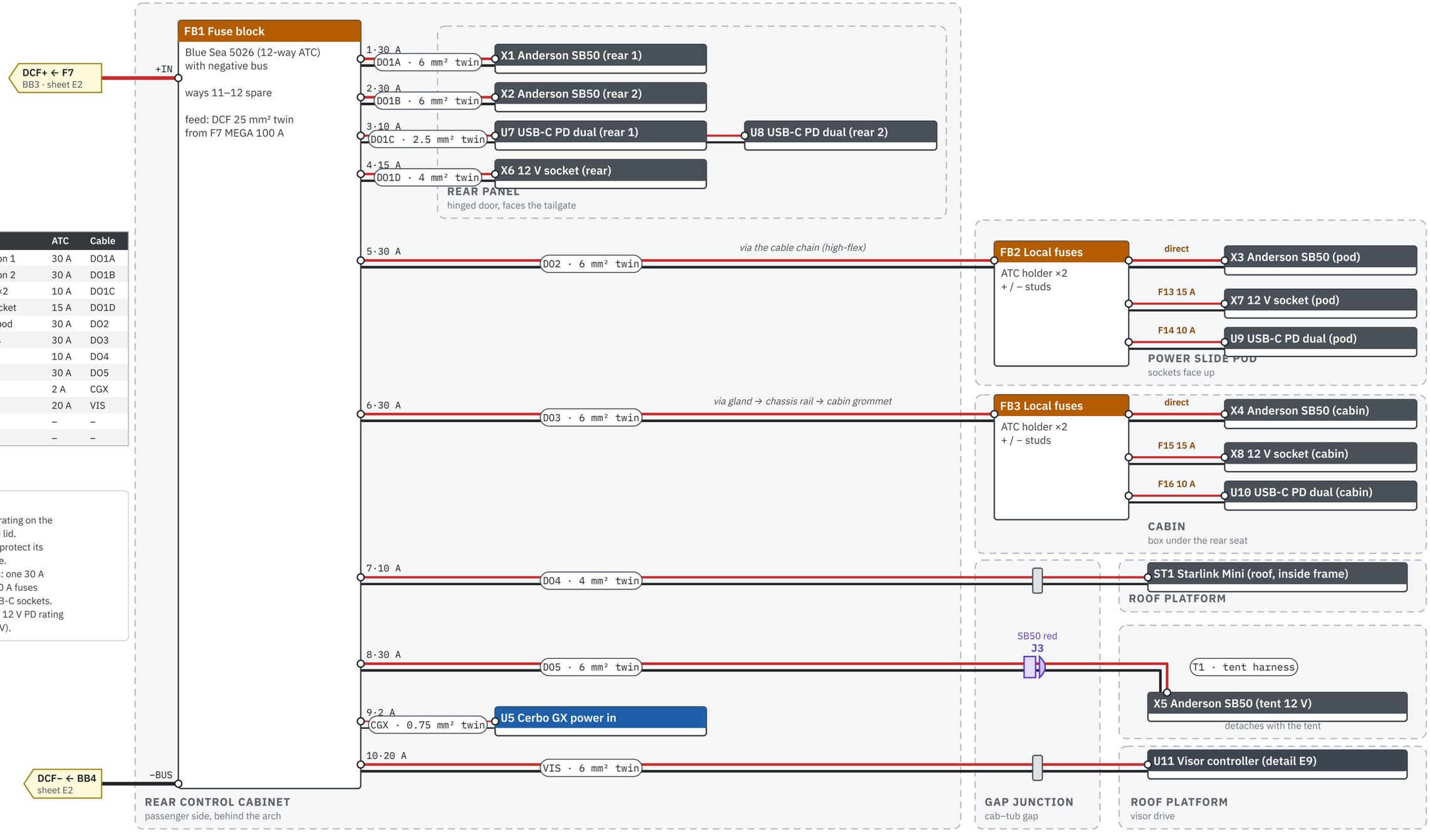
PROJECT: **Kia Tasman X-Pro fit-out · electrical**	SHEET: **E2 of E9**
TITLE: **DC power: battery bank and main distribution**	REV / DATE: **v0.11 · 2026-10-20**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.11.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

DC distribution: fuse block and 12 V outlets

FB1 in the rear control cabinet feeds every 12 V outlet. Twin cables: + from the numbered way, – to the FB1 negative bus.



FB1 WAYS

Way	Circuit	ATC	Cable
1	Rear Anderson 1	30 A	DO1A
2	Rear Anderson 2	30 A	DO1B
3	Rear USB-C ×2	10 A	DO1C
4	Rear 12 V socket	15 A	DO1D
5	Power slide pod	30 A	DO2
6	Cabin outlets	30 A	DO3
7	Starlink Mini	10 A	DO4
8	Tent 12 V	30 A	DO5
9	Cerbo GX	2 A	CGX
10	Visor drive	20 A	VIS
11	spare	–	–
12	spare	–	–

NOTES

- ATC blade fuses, rating on the way label and in the lid.
- Size each fuse to protect its cable, not the device.
- Pod / cabin boxes: one 30 A feed, local 15 A + 10 A fuses for the 12 V and USB-C sockets.
- USB-C: check the 12 V PD rating (many derate at 12 V).

LEGEND · line weight ∝ conductor size · `ID · size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) DC – (black)

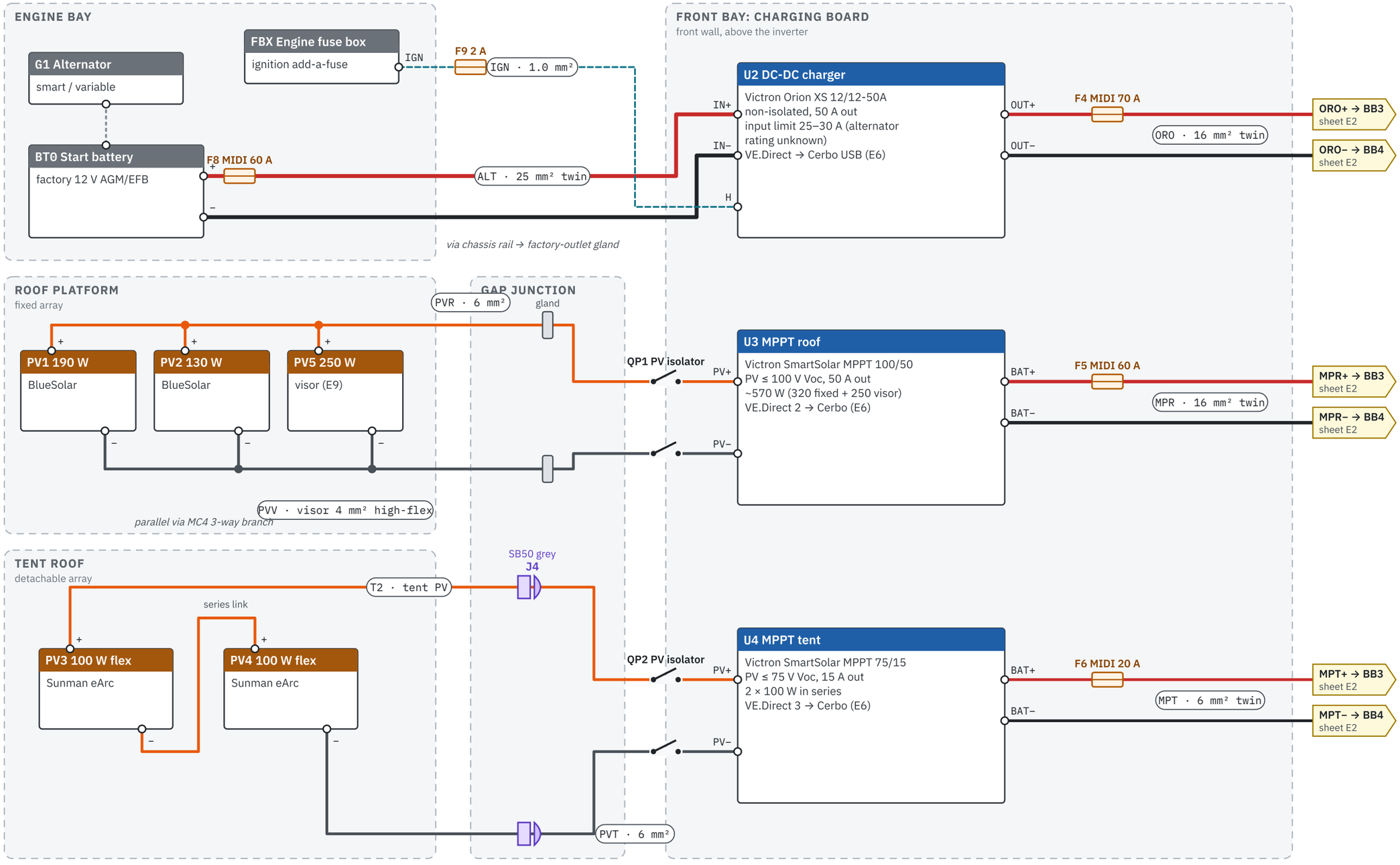
PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E3 of E9**
TITLE **DC distribution: fuse block and 12 V outlets**	REV / DATE **v0.11 · 2026-10-20**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.11.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

Charging: alternator (DC-DC) and solar

Three independent chargers on the front-wall board, each with its own MIDI fuse to the system busbars (sheet E2).



CHARGER SETTINGS

- Absorption / float / no temperature compensation set to iTechworld's $LiFePO_4$ figures in all three chargers (the BMS has no Victron link).
- Orion XS: charger mode, input current limit 25–30 A until the alternator rating is known; engine running detected by IGN on H.
- DVCC on with the SmartShunt as the battery monitor (Venus OS ≥ 3.20). QP1/QP2 are 2-pole DC isolators for servicing.

LEGEND · line weight ∝ conductor size · `ID` · `size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) · DC − (black) · PV + · PV − · Signal / sense · Factory vehicle wiring

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E4 of E9**
TITLE **Charging: alternator (DC-DC) and solar**	REV / DATE **v0.11 · 2026-10-20**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.11.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

230 V AC: shore inlet, inverter output, RCBOs and outlets

Licensed electrician to install and certify (AS/NZS 3000, AS/NZS 3001.2). Active brown · neutral blue · earth green/yellow.

XI1 Shore inlet
Clipsal 56AI315
IP66, 15 A

L N PE

Q4 RCBO
16 A 30 mA

T1 · AC-in cores

IP66 15 A
J2

tent: socket
vehicle: inlet

AC5 · 2.5 mm² 3-core

U1 MultiPlus-II 12/3000
AC-in → transfer switch (32 A)
→ AC-out 1
AC-in current limit: 15 A
Ground relay bonds N–PE while
inverting, so the RCBOs work
off-grid (check the AU setting).
AC-out 2 not used.

Case / PE → BB4 (PE1, E2)

CASE

PE1 → BB4 → CHB → chassis
sheet E2

FRONT BAY

RCB0 RCBO enclosure
IP66, DIN rail, N bar + PE bar

L N PE L N PE

Q1 10 A 30 mA → rear GPO

AC1 · 2.5 mm² 3-core

Q2 10 A 30 mA → pod GPO

Q3 10 A 30 mA → tent outlet

REAR CONTROL CABINET

L N PE

XS1 Rear double GPO
IP66, 10 A, rear panel

AC2 · 2.5 mm² 3-core

AC3 · 2.5 mm² 3-core

L N PE

XS2 Pod GPO
flush, faces up, 10 A

AC3: through the cable chain, high-flex 3-core

POWER SLIDE POD

XS3 Tent heater outlet
Clipsal 56SO310A
IP66, 10 A

PE N L

T1 · AC-out cores

vehicle: socket
tent: plug

J1
IP66 10 A

AC4 · 2.5 mm² 3-core

TENT SERVICES BOX
under the tent, front, inset

GAP JUNCTION
cab–tub gap

ELECTRICIAN NOTES

- Every 230 V circuit is RCD-protected (30 mA): Q4 on the inlet side, Q1–Q3 on AC-out.
- The tent has only two 230 V connections (J1 out, J2 in). They differ in rating and gender, so the tent's outlet can't be plugged into its own inlet.
- MultiPlus-II: AC-in limit 15 A; enable the ground relay for AU; test every RCD in inverter mode and on shore power before first use.
- Earth continuity: every metal enclosure and outlet earth back to U1 PE and BB4.
- Compliance plate and RCD test record inside the rear cabinet door.

LEGEND · line weight ∝ conductor size · `ID` · `size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

AC active (brown) · AC neutral (blue) · Protective earth (green/yellow)

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E5 of E9**
TITLE **230 V AC: shore inlet, inverter output, RCBOs and outlets**	REV / DATE **v0.11 · 2026-10-20**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.11.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

Control and communications

Cerbo GX gathers every Victron device for the GX Touch, Home Assistant and VRM. Hard-wired switches at the rear panel.

MONITORING
Cerbo GX in the rear cabinet

ST1 Starlink Mini
Wi-Fi router / uplink

Wi-Fi (VRM uplink, optional)

Wi-Fi

Victron integration · Modbus-TCP / MQTT

HA Home Assistant

U5 Cerbo GX MK2
Victron GX device, Venus OS ≥ 3.20
3 × VE.Direct, 2 × VE.Bus (RJ45)
3 × USB, HDMI, Ethernet, Wi-Fi, BLE
VE.Can 1/2, relays, tank / temp / digital inputs: unused

power: FB1 way 9, 2 A (CGX)

front bay / charging board

U6 GX Touch 50
5" touch display, rear panel

HDMI — D3 · HDMI+USB — USB 3

VE.Bus — D1 · VE.Bus RJ45 — VE.Bus → **U1 MultiPlus-II**

VE.Direct 1 — D2 · VE.Direct — VE.Direct → **P1 SmartShunt 500 A**

VE.Direct 2 — D2 · VE.Direct — VE.Direct → **U3 MPPT 100/50 (roof)**

VE.Direct 3 — D2 · VE.Direct — VE.Direct → **U4 MPPT 75/15 (tent)**

USB 1 — D4 · VE.Direct-USB — VE.Direct → **U2 Orion XS 12/12-50A**

CGX ← FB1 way 9 (2 A)
sheet E3 — PWR

RBF ← BB1 via F10 2 A
bank side, always live · E2

S1 Battery isolate switch
Blue Sea 2155 Contura
SPDT ON-ON with LED
rear panel

+12V
CTRL — SW1 · 4-core 1.0 mm² — red
LED — yel
GND — blk

QS1 ML-RBS 7713 control harness
red: CONTROL (high = close)
yellow: LED OUTPUT
black: GROUND
verify colours vs Blue Sea doc

→ BB4 system –
sheet E2

S2 Inverter on/off
SPST rocker, rear panel

1
2
SW2 · 2-core 0.75 mm²
L
H

U1 MultiPlus-II remote on/off
remove the factory L–H bridge

S3 Visor deploy / stow
momentary rocker + LED, rear panel

DEP
STOW
LED
VSW · 3-core 0.75 mm²

VSW → U11 visor controller
roof · sheet E9

IGN ← F9 2 A (E4) — H — **U2 Orion XS remote on/off**
H ← ignition 12 V (IGN)
remove the L–H bridge

VIG: ignition → U11 visor interlock
sheet E9

- S1 drives QS1 open/closed; QS1 latches, so S1 position is the command and the LED confirms the contacts.
- S2 open = MultiPlus fully off (inverter and charger).

Modes (on / charger only) are also on the GX Touch.

CONTROL CIRCUITS
switches on the rear panel

LEGEND · line weight ∝ conductor size · `ID` · `size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) · DC – (black) · Data (VE.Bus / VE.Direct / USB) · Control (12 V switched) · Signal / sense · Wireless (Wi-Fi / BLE)

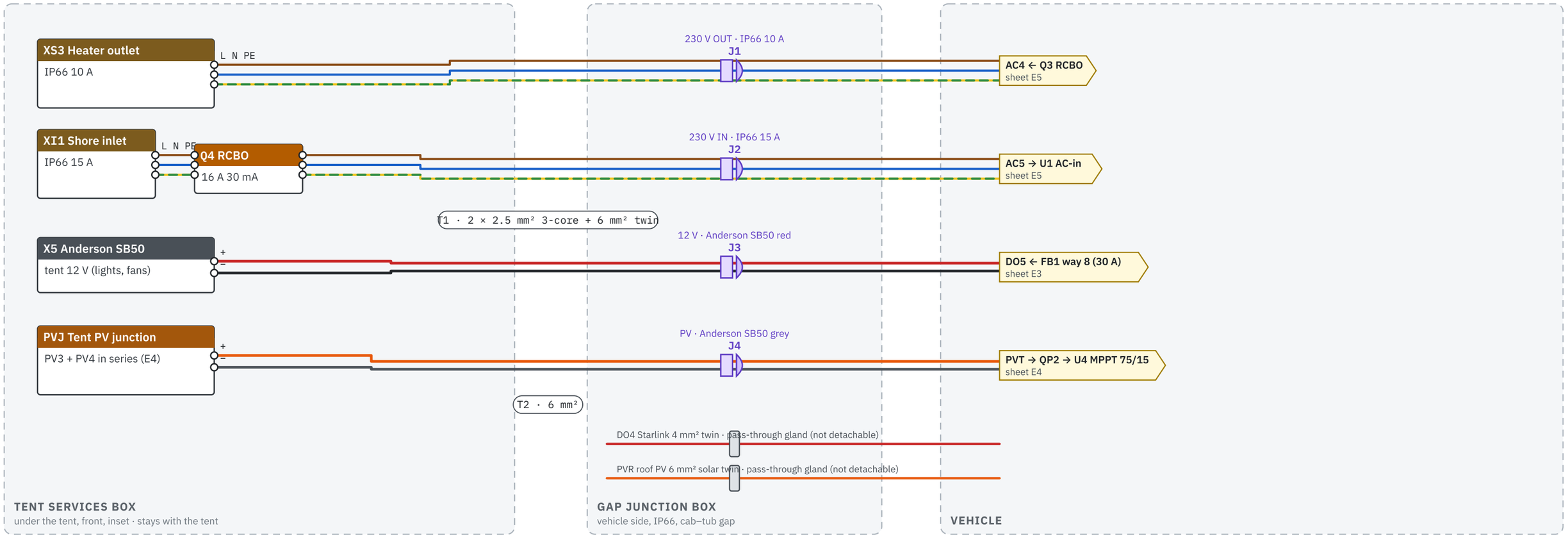
PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E6 of E9**
TITLE **Control and communications**	REV / DATE **v0.11 · 2026-10-20**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.11.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

Detachable tent harness and gap junction

Four keyed connectors at the cab–tub gap. Unplug J1–J4, cap them, and the tent lifts off with its harness and services box.



J1 · 230 V OUT TO TENT · IP66 10 A (CLIPSAL 56 SERIES)

Pin	Function	Conductor	Vehicle side	Tent side
A	Active	2.5 mm² brown	socket (live)	plug
N	Neutral	2.5 mm² blue		
E	Earth	2.5 mm² green/yellow		

J2 · 230 V SHORE IN · IP66 15 A INLET + EXTENSION SOCKET

Pin	Function	Conductor	Vehicle side	Tent side
A	Active (shore)	2.5 mm² brown	appliance inlet	socket (live)
N	Neutral	2.5 mm² blue		
E	Earth	2.5 mm² green/yellow		

J3 · 12 V · ANDERSON SB50 RED

Pin	Function	Conductor
+	12 V +	6 mm² red
–	12 V –	6 mm² black

J4 · PV · ANDERSON SB50 GREY

Pin	Function	Conductor
+	PV +	6 mm² solar
–	PV –	6 mm² solar

DETACH / REFIT

1. Switch the inverter off (S2) and isolate shore power at the source.
2. Unplug J2 (shore in) first, then J1, J3, J4.
3. Fit the caps to all eight halves (IP66 caps / SB50 dust covers).
4. Unclip the harness from the lid-front saddles, then lift the tent.

Refit in reverse. SB50 colours are keyed: red and grey won't mate, and J1/J2 differ in rating and gender, so nothing can cross-connect.

LEGEND · line weight ∝ conductor size · `ID` · `size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) · DC – (black) · PV + · PV – · AC active (brown) · AC neutral (blue) · Protective earth (green/yellow)

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E7 of E9**
TITLE **Detachable tent harness and gap junction**	REV / DATE **v0.11 · 2026-10-20**

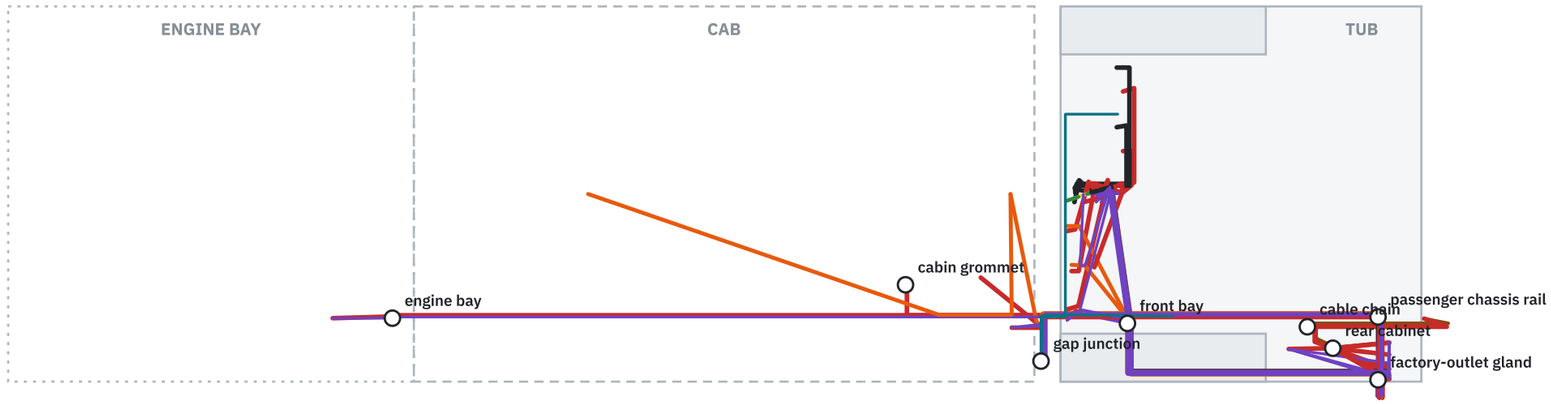
Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.11.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

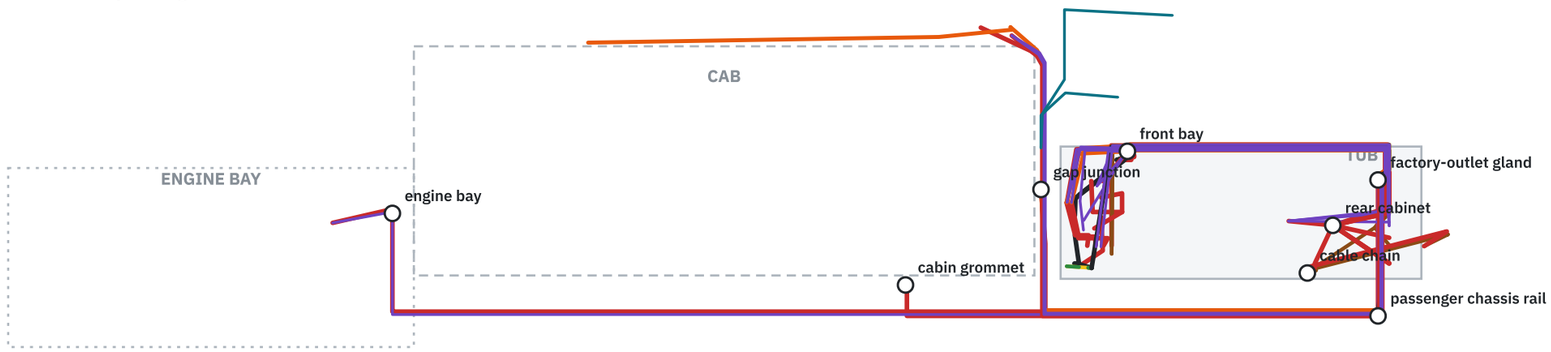
Cable routes: plan and side view

Generated from the routed 3D model (travel state). Every cable leaves the tub through the factory-outlet gland.

PLAN (from above, front of the vehicle to the left, driver side up)



SIDE (from the passenger side)



THROUGH THE FACTORY-OUTLET GLAND (13)
AC4, AC5, DO3, DO4, DO5, ALT, PVR, PVT, IGN, CHB, VIS, VSW
VIG

ALONG THE PASSENGER CHASSIS RAIL (12)
AC4, AC5, DO3, DO4, DO5, ALT, PVR, PVT, IGN, VIS, VSW, VIG

UP THE CAB–TUB GAP TO THE JUNCTION BOX (9)
AC4, AC5, DO4, DO5, PVR, PVT, VIS, VSW, VIG

IN THE POWER-SLIDE CABLE CHAIN (2)
AC3, DO2

DETACHABLE TENT HARNESS (2)
T1, T2

CABLE TO BUY (FROM THE ROUTED MODEL)

Cable	Model (m)	Buy (m)
0.75 mm2 red (supplied lead)	0.6	1
0.75 mm2 twin	0.5	1
1.0 mm2 red	0.5	1
1.0 mm2 single	15.0	18
1.5 mm2 twin	0.7	1
16 mm2 twin	2.0	3
2 x 2.5 mm2 3-core + 6 mm2 twin, sheathed bu	3.7	5
2 x 50 mm2 tinned, black	0.9	1
2 x 50 mm2 tinned, red	2.1	3
2-core 0.75 mm2	3.3	4
2.5 mm2 3-core flex	0.4	1
2.5 mm2 3-core flex (15 A)	6.2	8
2.5 mm2 3-core flex (TPS-flex)	3.2	4
2.5 mm2 3-core flex, UV rated	3.8	5
2.5 mm2 3-core high-flex (chain rated)	2.1	3
2.5 mm2 twin	0.6	1
25 mm2 black	4.0	5
25 mm2 twin (+/-), tinned	8.6	10
25 mm2 twin (red/black)	3.4	4
3-core 0.75 mm2 (deploy, stow, LED)	4.7	6
4 mm2 solar twin, high-flex (chain rated)	4.5	6
4 mm2 twin	0.6	1
4 mm2 twin + DC barrel lead	5.0	6
4-core 1.0 mm2 (+12 V feed, control, LED, gr	3.0	4
50 mm2 green/yellow	0.4	1
6 mm2 solar twin (UV)	14.3	17
6 mm2 twin	14.9	18
6 mm2 twin high-flex	2.0	3
70 mm2 tinned, black	3.1	4
70 mm2 tinned, red	2.4	3
GX Touch cable (HDMI + USB)	0.8	1
RJ45 VE.Bus cable	3.5	5
VE.Direct cables x3	4.1	5
VE.Direct to USB interface	3.9	5

Scale ≈ 1 : 26 on A3 · 50 cables · total 125.3 m (model + 10 % + 0.3 m per cable)

LEGEND · line weight ∝ conductor size · `ID` · `size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) · DC – (black) · PV + · AC active (brown) · Protective earth (green/yellow) · Data (VE.Bus / VE.Direct / USB) · Control (12 V switched)

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E8 of E9**
TITLE **Cable routes: plan and side view**	REV / DATE **v0.11 · 2026-10-20**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.11.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

Visor drive: slide, tilt and locking

The windscreen visor panel (PV5) slides out on two rails and tilts to 20°. One button at the rear panel; every motion is interlocked.

FEEDS
rear cabinet · E3, E6

VIS+ ← FB1 way 10, ATC 20 A
sheet E3

VIS− ← FB1 neg bus
sheet E3

VIS · 6 mm² twin

+12V
GND

DEP ← S3
STOW ← S3
LED → S3

S3 at the rear panel, sheet E6

VSW · 3-core 0.75 mm²

DEP
STOW
LED

VIG ← Orion H (ignition)
sheet E6

VIG · 1.0 mm²

IGN

INPUTS
S3 switches DEP / STOW to ground;
U11 pulls the inputs up.
LED: flashing = moving,
solid = not stowed, off = locked.
IGN high = engine on.

U11 Visor controller
IP66 enclosure, platform front
12 V logic (PLC / µC) + 2 × dual H-bridge drivers, current sense
3 motor channels: M2, M3, M4
latch driver: Y1 + Y2
4 limit inputs: B1–B4
BZ1 buzzer on the lid

motion only while the button is held (dead-man), 1 s to start
timeout 25 s per step → fault

manual override: 8 mm hex on M2 shaft end and both tilt drives

M2+
M2−
M3+
M3−
M4+
M4−
Y+
Y−
IN1
IN2
IN3
IN4
COM

ROOF PLATFORM
visor controller, front cross-member

M2 Slide gearmotor
12 V worm gearmotor, self-locking
cross shaft → 2 × HTD belts

local · 2.5 mm²

M3 Tilt drive, driver side
12 V worm rotary actuator
at the hinge beam end

1.5 mm² in chain

M4 Tilt drive, passenger side
12 V worm rotary actuator
runs in step with M3

1.5 mm² in chain

Y1/Y2 Front latches (×2, parallel)
spring-locked, 12 V to release
power lost = locked

IP67 roller-lever switches, NO; B4 is set 1° before the hard stop

B1 limit · carriages home (stowed)
B2 limit · carriages out (deployed)
B3 limit · tilt at 0° (flat)
B4 limit · tilt at 21° (stop at 22°)

COM

VISOR MECHANISM
rails, carriages, hinge beam

PV5 250 W visor
semi-flex on honeycomb

PVV · 4 mm²

PVV → MC4 3-way → PVR → U3
MPPT 100/50 · sheet E4

high-flex, runs in an igus chain beside the left rail

VISOR PANEL
moves with the carriages

SEQUENCE

Step	Deploy (hold DEP)	Stow (hold STOW)
1	check IGN off, B1 + B3 closed	allowed any time (IGN on too)
2	Y1/Y2 energise (pins out)	M3 + M4 tilt down until B3
3	M2 extends until B2	Y1/Y2 energise; M2 retracts to B1
4	M3 + M4 tilt up to 20°, B4 = limit	Y1/Y2 off: springs lock the pins
5	Y1/Y2 off; worms hold the panel	LED off = locked for travel

INTERLOCKS AND SAFETY

- No deploy with the ignition on. Engine on while not stowed → BZ1 sounds, LED flashes; nothing moves without the button.
- M2 runs only with B3 closed: the panel slides flat, then tilts.
- Over-current on any channel → stop and back off 2 mm (pinch).
- Mechanical stop at 22° keeps ≥ 87 mm to the glass even if B4 fails; worm drives cannot be back-driven by wind or load.
- Latches fail locked. Stow when gusts exceed ~40 km/h.
- U11 draws < 20 mA idle; isolate with FB1 way 10.

LEGEND · line weight ∝ conductor size · ID · size tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) · DC − (black) · PV + · PV − · Control (12 V switched) · Signal / sense

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E9 of E9**
TITLE **Visor drive: slide, tilt and locking**	REV / DATE **v0.11 · 2026-10-20**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.11.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

Schedules

Cable schedule

Lengths are measured along each cable's routed path in the 3D model, plus 10 % slack and 0.3 m for terminations.

ID	CIRCUIT	FROM	TO	CABLE	PROTECTION	M	HARNESS
B1P	Bank +	Battery 1 + (MRBF 300 A)	Bank + busbar	70 mm² tinned, red	MRBF 300 A on terminal	0.55	vehicle
B1N	Bank -	Battery 1 -	Bank - busbar	70 mm² tinned, black	-	0.93	vehicle
B2P	Bank +	Battery 2 + (MRBF 300 A)	Bank + busbar	70 mm² tinned, red	MRBF 300 A on terminal	0.83	vehicle
B2N	Bank -	Battery 2 -	Bank - busbar	70 mm² tinned, black	-	1.21	vehicle
RB1	Main +	Bank + busbar	Remote battery switch IN	70 mm² tinned, red	MRBF x2 upstream	0.44	vehicle
RB2	Main +	Remote battery switch OUT	System + busbar	70 mm² tinned, red	-	0.59	vehicle
SH1	Main -	Bank - busbar	SmartShunt (battery side)	70 mm² tinned, black	-	0.38	vehicle
SH2	Main -	SmartShunt (system side)	System - busbar	70 mm² tinned, black	-	0.54	vehicle
CT1	Inverter +	System + busbar	Class-T 400 A	2 x 50 mm² tinned, red	-	0.94	vehicle
INVP	Inverter +	Class-T 400 A	MultiPlus-II DC +	2 x 50 mm² tinned, red	Class-T 400 A	1.14	vehicle
INVN	Inverter -	System - busbar	MultiPlus-II DC -	2 x 50 mm² tinned, black	-	0.86	vehicle
ORO	Charging output	orion xs 50a	System +/- busbars (MIDI fuse)	16 mm² twin	MIDI 70 A	1.11	vehicle
MPR	Charging output	mppt roof 100 30	System +/- busbars (MIDI fuse)	16 mm² twin	MIDI 60 A	0.93	vehicle
MPT	Charging output	mppt tent 75 15	System +/- busbars (MIDI fuse)	6 mm² twin	MIDI 20 A	0.75	vehicle
DCF	DC distribution	System + busbar (MEGA 100 A)	Rear DC fuse block	25 mm² twin (red/black)	MEGA 100 A	3.36	vehicle
AC1	AC out	MultiPlus-II AC-out 1	RCBO enclosure	2.5 mm² 3-core flex (TPS-flex)	MultiPlus + RCBOs	3.21	vehicle
AC2	Rear GPO	RCBO 1	Rear panel double GPO	2.5 mm² 3-core flex	RCBO 10 A 30 mA	0.4	vehicle
AC3	Slide GPO	RCBO 2	Power-slide pod GPO	2.5 mm² 3-core high-flex (chain rated)	RCBO 10 A 30 mA	2.13	vehicle
AC4	Tent heater GPO	RCBO 3	Gap junction -> tent services box GPO	2.5 mm² 3-core flex, UV rated	RCBO 10 A 30 mA	3.79	vehicle
AC5	Shore inlet	Gap junction (from tent 15 A inlet)	MultiPlus-II AC-in	2.5 mm² 3-core flex (15 A)	RCBO 16 A 30 mA in the services box + MultiPlus AC-in limit 15 A	6.15	vehicle
DO1A	Rear Anderson 1	DC fuse block	Rear panel Anderson 1	6 mm² twin	ATC 30 A	0.63	vehicle
DO1B	Rear Anderson 2	DC fuse block	Rear panel Anderson 2	6 mm² twin	ATC 30 A	0.62	vehicle
DO1C	Rear USB-C x2	DC fuse block	Rear panel USB-C x2	2.5 mm² twin	ATC 10 A	0.57	vehicle
DO1D	Rear 12 V socket	DC fuse block	Rear panel 12 V socket	4 mm² twin	ATC 15 A	0.61	vehicle

ID	CIRCUIT	FROM	TO	CABLE	PROTECTION	M	HARNESS
DO2	Slide DC	DC fuse block	Power-slide pod Anderson / 12 V / USB-C	6 mm² twin high-flex	ATC 30 A (+ inline 15 A / 10 A in the pod)	2.0	vehicle
DO3	Cabin outlets	DC fuse block	Under-rear-seat box (Anderson, USB-C, 12 V)	6 mm² twin	ATC 30 A (+ inline 15 A / 10 A in the box)	4.18	vehicle
DO4	Starlink Mini	DC fuse block	Starlink Mini (roof)	4 mm² twin + DC barrel lead	ATC 10 A	5.0	vehicle
DO5	Tent 12 V	DC fuse block	Gap junction -> tent (Anderson SB50)	6 mm² twin	ATC 30 A	3.98	vehicle
ALT	Alternator charge	Start battery (60 A MIDI)	Orion XS input	25 mm² twin (+/-), tinned	60 A at start battery	8.62	vehicle
PVR	Roof PV	Roof panels + visor (parallel, MC4 3-way branch)	MPPT 100/50 PV in	6 mm² solar twin (UV)	2-pole DC PV isolator 32 A	6.9	vehicle
PVT	Tent PV	Gap junction (from tent panels)	MPPT 75/15 PV in	6 mm² solar twin (UV)	2-pole DC PV isolator 32 A	5.79	vehicle
SW1	Battery switch control	Rear panel ML-RBS switch (Blue Sea 2155)	ML-RBS control harness	4-core 1.0 mm² (+12 V feed, control, LED, ground)	2 A inline at bank + (RBF)	2.98	vehicle
RBF	Battery switch control	Bank + busbar	ML-RBS control feed (inline 2 A)	1.0 mm² red	ATC 2 A inline	0.47	vehicle
SHV	Shunt sense	Bank + busbar	SmartShunt Vbatt+	0.75 mm² red (supplied lead)	1 A inline (supplied)	0.64	vehicle
FAN	Inverter fan	System + busbar	92 mm thermostatic fan	1.5 mm² twin	ATC 5 A inline	0.67	vehicle
CGX	Cerbo GX power	DC fuse block	Cerbo GX power in	0.75 mm² twin	ATC 2 A	0.5	vehicle
IGN	Orion ignition sense	Engine-bay fuse box (add-a-fuse, ignition)	Orion XS remote H	1.0 mm² single	ATC 2 A add-a-fuse	8.64	vehicle
PE1	Inverter case earth	MultiPlus-II chassis earth stud	System - busbar	50 mm² green/yellow	-	0.42	vehicle
CHB	Chassis bond	System - busbar	Chassis rail bond point	25 mm² black	-	4.02	vehicle
SW2	Inverter remote on/off	Rear panel switch	MultiPlus-II remote terminals	2-core 0.75 mm²	-	3.26	vehicle
D1	VE.Bus	Cerbo GX	MultiPlus-II	RJ45 VE.Bus cable	-	3.55	vehicle
D2	VE.Direct	Cerbo GX VE.Direct 1-3	SmartShunt / MPPT 100/50 / MPPT 75/15	VE.Direct cables x3	-	4.08	vehicle
D4	VE.Direct	Cerbo GX USB	Orion XS VE.Direct	VE.Direct to USB interface	-	3.92	vehicle
D3	Display	Cerbo GX (HDMI + USB)	GX Touch 50	GX Touch cable (HDMI + USB)	-	0.77	vehicle
PVV	Visor PV	Visor panel (moving)	Roof PV branch (parallel with PV1/PV2)	4 mm² solar twin, high-flex (chain rated)	-	4.53	vehicle
VIS	Visor drive power	DC fuse block (way 10)	Visor controller (roof)	6 mm² twin	ATC 20 A	4.77	vehicle
VSW	Visor control	Rear panel visor button (S3)	Visor controller (roof)	3-core 0.75 mm² (deploy, stow, LED)	-	4.67	vehicle
VIG	Visor interlock	Orion XS remote H (ignition, F9)	Visor controller interlock input	1.0 mm² single	F9 2 A (shared)	6.34	vehicle
T1	Tent harness	Tent services box	Gap junction plugs (240 V out, 240 V in, 12 V)	2 x 2.5 mm² 3-core + 6 mm² twin, sheathed bundle	RCBO 16 A 30 mA (AC-in, in the services box)	3.74	tent
T2	Tent PV harness	Tent panels	Gap junction (PV plug)	6 mm² solar twin (UV)	-	1.56	tent

Protection

REF	DEVICE	RATING	LOCATION	PROTECTS
F1	MRBF terminal fuse (Blue Sea 5191)	300 A	Battery 1 + stud	B1P
F2	MRBF terminal fuse (Blue Sea 5191)	300 A	Battery 2 + stud	B2P
F3	Class-T fuse + holder (Blue Sea 5502)	400 A	BB3 system + bus	CT1, INVP (MultiPlus DC)
F4	MIDI fuse	70 A	BB3 stud	ORO (Orion XS output)
F5	MIDI fuse	60 A	BB3 stud	MPR (MPPT 100/50 output)
F6	MIDI fuse	20 A	BB3 stud	MPT (MPPT 75/15 output)
F7	MEGA fuse	100 A	BB3 stud	DCF (feed to FB1)
F8	MIDI fuse + holder	60 A	Start battery +, engine bay	ALT
F9	Add-a-fuse, ignition circuit	2 A	Engine-bay fuse box	IGN
F10	Inline ATC holder	2 A	BB1 bank + bus	RBF, SW1 (ML-RBS control)
F11	Inline fuse (supplied with SmartShunt)	1 A	BB1 bank + bus	SHV
F12	Inline ATC holder	5 A	BB3 stud	FAN
FB1 1–10	ATC blade fuses in FB1 (Blue Sea 5026)	see E3	Rear control cabinet	DO1A–DO5, CGX, VIS (way 10, 20 A)
F13 / F14	ATC in FB2	15 A / 10 A	Power slide pod	pod 12 V socket / USB-C
F15 / F16	ATC in FB3	15 A / 10 A	Cabin box	cabin 12 V socket / USB-C
Q1–Q3	RCBO 1P+N, 30 mA	10 A	RCBO enclosure, rear cabinet	AC2, AC3, AC4
Q4	RCBO 1P+N, 30 mA	16 A	Tent services box	shore inlet → AC-in (T1, AC5)
QS1	ML-RBS remote battery switch (Blue Sea 7713)	500 A	Front bay	isolation, bank + → system +
QP1 / QP2	2-pole DC PV isolator	32 A	Front bay, beside the MPPTs	PVR / PVT (servicing)

FB1 fuse block ways

WAY	CIRCUIT	ATC	CABLE
1	Rear Anderson 1	30 A	DO1A
2	Rear Anderson 2	30 A	DO1B
3	Rear USB-C ×2	10 A	DO1C
4	Rear 12 V socket	15 A	DO1D
5	Power slide pod	30 A	DO2
6	Cabin outlets	30 A	DO3
7	Starlink Mini	10 A	DO4
8	Tent 12 V	30 A	DO5
9	Cerbo GX	2 A	CGX
10	Visor drive	20 A	VIS
11	spare	–	–
12	spare	–	–

Before power-up

- Measure the real cable runs: these lengths come from [EST] positions in the model.
- Confirm the ML-RBS control-wire colours and switch wiring against the Blue Sea 7713 instructions.
- Set every charger to iTechworld's $LiFePO_4$ voltages; set the Orion XS input limit to 25–30 A.
- Torque every stud, then re-check after the first corrugated road.
- Electrician: earth continuity, polarity, RCD trip tests on shore power and in inverter mode.

Component register

REF	ITEM	MODEL	LOCATION	SHEETS
BT1, BT2	House batteries 12.8 V 400 Ah	iTechworld iTech400X Pro	Front bay, on end	E2
BB1–BB4	Busbars (bank ±, system ±)	Blue Sea PowerBar 600 A	Front bay distribution column	E2

REF	ITEM	MODEL	LOCATION	SHEETS
QS1	Remote battery switch	Blue Sea ML-RBS 7713 (12 V)	Front bay	E2, E6
P1	Battery monitor	Victron SmartShunt 500 A	Front bay	E2, E6
U1	Inverter/charger	Victron MultiPlus-II 12/3000/120-32 230 V	Front bay, lying flat	E2, E5, E6
U2	DC-DC charger	Victron Orion XS 12/12-50A	Charging board	E4, E6
U3	Solar charger, roof	Victron SmartSolar MPPT 100/50	Charging board	E4, E6
U4	Solar charger, tent	Victron SmartSolar MPPT 75/15	Charging board	E4, E6
M1	Inverter fan	92 mm 12 V thermostatic	Above the inverter	E2
FB1	Fuse block 12-way + neg bus	Blue Sea 5026	Rear control cabinet	E3
FB2, FB3	Local fuse holders + studs	2-way ATC	Power slide pod / cabin box	E3
U5	GX device	Victron Cerbo GX MK2	Rear control cabinet	E3, E6
U6	Display	Victron GX Touch 50	Rear panel	E6
S1	Battery isolate switch	Blue Sea 2155 Contura SPDT with LED	Rear panel	E6
S2	Inverter on/off switch	SPST rocker	Rear panel	E6
S3	Visor deploy / stow button	momentary ON-OFF-ON rocker with LED	Rear panel	E6, E9
RCBO	RCBO enclosure, Q1–Q3	IP66 DIN enclosure, 3 × RCBO 10 A 30 mA	Rear control cabinet	E5
XS1	Rear double GPO	IP66 10 A (Clipsal 56 series)	Rear panel	E5
XS2	Pod GPO	Flush 10 A, faces up	Power slide pod	E5
XS3	Tent heater outlet	Clipsal 56SO310A IP66 10 A	Tent services box	E5, E7
XI1, Q4	Shore inlet + RCBO	Clipsal 56AI315 15 A + RCBO 16 A 30 mA	Tent services box	E5, E7
X1–X5	Anderson SB50 sockets	Anderson SB50 red + flush housing	Rear panel ×2, pod, cabin, tent box	E3
X6–X8	12 V sockets	Narva / Blue Sea	Rear panel, pod, cabin	E3
U7–U10	Dual USB-C PD sockets	check the 12 V PD rating	Rear panel ×2, pod, cabin	E3
J1	230 V out connector	IP66 10 A socket (vehicle) / plug (tent)	Gap junction box	E5, E7
J2	230 V in connector	IP66 15 A inlet (vehicle) / extension socket (tent)	Gap junction box	E5, E7
J3	12 V tent connector	Anderson SB50 red	Gap junction box	E3, E7
J4	Tent PV connector	Anderson SB50 grey	Gap junction box	E4, E7
PV1, PV2	Roof panels 190 W + 130 W	Victron BlueSolar (parallel)	Roof platform	E4
PV5	Visor panel ~250 W	semi-flexible on honeycomb backing (parallel with PV1/PV2)	Slide-out visor	E4, E9
U11	Visor controller	12 V PLC / µC + 2 × dual H-bridge, IP66 box	Roof platform, front	E3, E9
M2	Slide gearmotor	12 V worm gearmotor, self-locking	Roof platform, rear cross-member	E9
M3, M4	Tilt drives	12 V worm rotary actuators	Hinge beam ends	E9
Y1, Y2	Front latches	12 V solenoid, spring-locked (fail-locked)	Platform front	E9

REF	ITEM	MODEL	LOCATION	SHEETS
B1–B4	Limit switches	IP67 roller-lever, NO	Rails and hinge beam	E9
BZ1	Not-stowed buzzer	12 V piezo	Visor controller lid	E9
PV3, PV4	Tent panels 2 × 100 W	Sunman eArc flexible (series)	Tent roof	E4
ST1	Satellite internet	Starlink Mini	Roof platform, inside the frame	E3, E6
BT0, G1, FBX	Vehicle start battery, alternator, fuse box	factory	Engine bay	E4

Generated by `tools/wiring/diagram.py` from `cad/fitout/wiring.py` and the routed 3D model. Rebuild: `cd cad && ../.venv/bin/python build.py && cd .. && python3 tools/wiring/diagram.py`.